

Chapter 3

Functions

You
will
be
fa-
mil-
iar
with
the
no-
tion
of
a
func-
tion
in
math-
e-
mat-
ics
and
have
used
them
many
times
up
to
now.
But
in
this
chap-
ter
we
just
wish
to
for-
malise
the

CHAPTER
3.
FUNCTIONS

def-
i-
ni-
tion
of
a
func-
tion
and
in-
tro-
duce
var-
i-
ous
prop-
er-
ties
of
func-
tions.
Through-
out
this
we
wish
to
em-
pha-
size
the
treat-
ment
of
func-
tions
as
math-
e-
mat-
i-
cal
ob-
jects
in
them-
selves.

Definition

3.1
(Func-
tion).
A
func-
tion
f

from
a
set
 X
to
a
set
 Y
is
a
rule
which
as-
signs
el-
e-
ments
of
the
set
 Y
to
el-
e-
ments
of
 X .
It
must
do
this
in
such
a
way
that
to
each
el-
e-
ment
of
 X
there
is
one
and
only
one
el-
e-
ment
of
 Y
as-
so-

CHAPTER
3.
FUNCTIONS
ci-
ated
to
it.

This
con-
di-
tion
is
some-
times
re-
ferred
to
as
say-
ing
that
func-
tions
are
*single-
valued*.

Definition

3.2
(Func-
tion
ter-
mi-
nol-
ogy
and
no-
ta-
tion).
We
can
write

$$f : X \rightarrow Y$$

to
de-
note
that
 f
is
a
func-
tion
from
the
set
 X
to

the
set
 Y .
The
set
 X
is
called
the
do-
main
of
 f
and
 Y
is
called
the
co-
domain
of
 f .
An
el-
e-
ment
 $x \in$
 X
is
called
an
ar-
gu-
ment
of
the
func-
tion
 f
and
we
use
 $f(x)$
to
de-
note
the
as-
so-
ci-
ated
el-
e-
ment
of
 Y ,

CHAPTER
3.
FUNCTIONS
called
the
value
of
the
func-
tion
 f
at
the
ar-
gu-
ment
 x .

Note
that
there
is
no
re-
quire-
ment
that
all
the
el-
e-
ments
of
the
co-
domain
oc-
cur
as
val-
ues
of
the
func-
tion
but
there
is
a
re-
quire-
ment
that
a
value
 $f(x)$
is
de-
fined

for
 ev-
 ery
 el-
 e-
 men-
 t/ar-
 gu-
 ment
 x
 from
 the
 do-
 main
 of
 f .

Example

3.1.

As
 men-
 tioned
 above
 you
 will
 be
 fa-
 mil-
 iar
 with
 these
 con-
 cepts
 from
 your
 study
 of
 math-
 e-
 mat-
 ics
 up
 to
 now.

1. The
 square
 root
 func-
 tion
 $s :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}$
 is
 the
 func-

CHAPTER
3.
FUNCTIONS

tion
from
the
do-
main
 $\mathbb{R}^+$
of
pos-
i-
tive
real
num-
bers
to
the
co-
domain
 $\mathbb{R}$,
de-
fined
by
the
for-
mula

for all $x \in \mathbb{R}^+$, $s(x) = \sqrt{x}$.

2.

The
trigono-
met-
ric
sine
func-
tion,
 $\sin :$
 $\mathbb{R} \rightarrow$
 $[-1, 1]$.
This
can
be
de-
fined
in
many
ways.
One
way
is
to
de-
fine
 $\sin(\theta)$
as
the
 y -
coordinate

of
the
point
on
the
unit
cir-
cle
in
the
plane
which
makes
an
an-
gle
of
 θ
ra-
di-
ans
with
the
pos-
i-
tive
di-
rec-
tion
of
the
 x -
axis,
see
fig-
ure
[3.1](#).

Definition

3.3

(Range
/
im-
age).
The
range
or
*im-
age*
of
a
func-
tion
 $g :$
 $V \rightarrow$
 W

CHAPTER
3.
FUNCTIONS

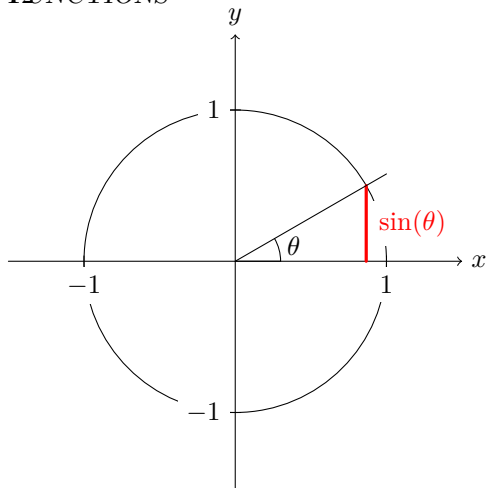


Figure 3.1:

A
def-
i-
ni-
tion
of
 $\sin(\theta)$.

is
the
sub-
set
of
the
co-
domain
 W
con-
sist-
ing
of
the
val-
ues
of
 g ,
i.e.

$$\text{im}(g) = \{ g(v) : v \in V \}.$$

When
dis-
cussing
par-
tic-
u-
lar

func-
tions
it
is
of-
ten
the
case
that
the
do-
main
and
co-
domain
will
not
be
ex-
plic-
itly
men-
tioned.
The
do-
main
for
the
func-
tion
should
be
clear
from
the
con-
text,
i.e.
to
what
set
of
ar-
gu-
ments
is
the
func-
tion
in-
tended
to
ap-
ply.
The
pre-

CHAPTER
3.
FUNCTIONS
cise
de-
scrip-
tion
of
the
co-
domain
is
a
bit
trick-
ier
as
the
choice
of
the
co-
domain
is
some-
what
ar-
bi-
trary
as
long
as
it
con-
tains
the
im-
age
of
the
func-
tion.
How-
ever
for
some
prop-
er-
ties
and
op-
er-
a-
tion
such
as
sur-
jec-
tiv-

ity
 and
com-
po-
si-
tion
 some
 con-
 sid-
 er-
 a-
 tion
 will
 need
 to
 be
 given
 to
 the
 ap-
 pro-
 pri-
 ate
 co-
 domain
 of
 the
 func-
 tion.

Definition

3.4

(In-
 jec-
 tive
 /
 one-
 to-one).

A
 func-
 tion

$g :$
 $U \rightarrow$
 V

is
in-
jec-
tive

if
 and
 only

if
 no
 two
 el-
 e-

CHAPTER
3.
FUNCTIONS

ments
from
the
do-
main
 U
get
mapped
to
the
same
value
in
the
co-
domain
 V .
This
con-
di-
tion
can
be
ex-
pressed
for-
mally
as:

$$\forall u_1, u_2 \in U \quad g(u_1) = g(u_2) \Rightarrow u_1 = u_2.$$

A
more
in-
for-
mal,
but
de-
scrip-
tive,
term
for
this
prop-
erty
is
to
say
that
such
a
func-
tion
 g
is
one-

*to-
one.*

Definition

3.5

(Sur-
jec-
tive
/
onto).

A
func-
tion

$g :$
 $U \rightarrow$
 V

is
*sur-
jec-
tive*

if
and
only

if
ev-
ery
el-
e-
ment

of
the
do-
main

oc-
curs

as
a
value

of
 g .

Put
more

for-
mally

this
con-
di-

tion
is:

$\forall v \in V \exists u \in U$ such that $g(u) = v$.

A
more
in-
for-
mal,

CHAPTER
3.
FUNCTIONS
but
de-
scrip-
tive,
term
for
this
prop-
erty
is
to
say
that
such
a
func-
tion
 g
is
onto.

These
prop-
er-
ties
can
of-
ten
de-
pend
on
the
ex-
act
choice
of
the
do-
main
and
co-
domain,
as
the
fol-
low-
ing
ex-
am-
ples
show.

Example
3.2.
The
func-
tion

$f_1 :$
 $(0, \frac{\pi}{2}) \rightarrow$
 $(0, 1)$
 de-
 fined
 by

$$f_1(t) = \sin t,$$

is
 both
 in-
 jec-
 tive
 and
 sur-
 jec-
 tive,
 i.e.
 bi-
 jec-
 tive.

However

the
 func-
 tion
 $f_2 :$
 $(0, \pi) \rightarrow$
 $(0, 2)$
 de-
 fined
 by

$$f_2(t) = \sin t,$$

is
 nei-
 ther
 in-
 jec-
 tive
 nor
 sur-
 jec-
 tive.

Since

$$\sin\left(\frac{\pi}{4}\right) =$$

$$\sin\left(\frac{3\pi}{4}\right) =$$

$$\frac{1}{\sqrt{2}},$$

f_2
 is
 not
 in-
 jec-
 tive
 and
 the

CHAPTER
 3.
 FUNCTIONS
 fact
 that
 for
 all
 t ,
 $|\sin t| \leq$
 1
 means
 that
 there
 is
 no
 t
 for
 which
 $\sin t =$
 1.5,
 and
 so
 f_2
 is
 not
 sur-
 jec-
 tive.

Definition

3.6
 (Bi-
 jec-
 tive).
 A
 func-
 tion
 is
bi-
jec-
tive
 if
 and
 only
 if
 it
 is
 both
 in-
 jec-
 tive
 and
 sur-
 jec-
 tive.

3.1. REAL-VALUED FUNCTIONS

3.1 Real-valued functions

A lot of the time the functions we encounter in mathematics are *real-valued*, i.e. functions whose domains and codomains are subsets of the real numbers $\mathbb{R}$.

Algebra of real-valued functions

Using the familiar

CHAPTER
3.
FUNCTIONS
op-
er-
a-
tions
of
the
real
num-
bers
we
can
de-
fine
as-
so-
ci-
ated
op-
er-
a-
tions
amongst
real-
valued
func-
tions.
We
want
to
stress
here
the
con-
sid-
er-
a-
tion
of
func-
tions
as
math-
e-
mat-
i-
cal
ob-
jects
in
their
own
right
with
op-
er-
a-

3.1.
**REAL-
 VALUED
 FUNCTIONS**
 have
 de-
 fined
 on
 them.

Definition

3.7

(Ad-
 di-
 tion).
 Given
 two
 real-
 valued
 func-
 tions
 f
 and
 g
 with
 the
 same
 do-
 main
 we
 can
 de-
 fined
 a
 new
 func-
 tion
 $f +$
 g ,
 called
 the
sum
 of
 f
 and
 g ,
 on
 this
 do-
 main
 us-
 ing
 the
 for-
 mula

$$(f + g)(x) = f(x) + g(x).$$

Definition

3.8

(Scalar

CHAPTER
3.
FUNCTIONS

mul-
ti-
pli-
ca-
tion).

Let
 f
be
a
real-
valued
func-
tion
and
 $\lambda \in$
 $\mathbb{R}$.

Then
we
can
form
a
new
func-
tion
 λf ,
which
has
the
same
do-
main
as
 f ,
and
is
de-
fined
by
the
for-
mula

$$(\lambda f)(x) = \lambda(f(x)).$$

Definition

3.9
(Mul-
ti-
pli-
ca-
tion).

Let
 f
and
 g
be
two

3.1. REAL- VALUED FUNCTIONS

real-
valued
func-
tions
with
the
same
do-
main.
Then

we
can
form
a
new
func-
tion

fg ,
on
the
same
do-
main,
de-
fined
by
the
for-
mula

$$(fg)(x) = f(x)g(x).$$

Definition

3.10

(Re-
cip-
ro-
cal).

Let

f
be

a
real-
valued
func-
tion

that
never
takes
the
value

zero.
Then

we
can
form
a

CHAPTER
3.
FUNCTIONS

new
func-
tion
 $\frac{1}{f}$,
on
the
same
do-
main
as
 f ,
de-
fined
by
the
for-
mula

$$\left(\frac{1}{f}\right)(x) = \frac{1}{f(x)}.$$

Note
that
we
do
not
like
to
use
the
no-
ta-
tion
 f^{-1}
for
the
re-
cip-
ro-
cal
func-
tion
as
this
is
nor-
mally
re-
served
for
the
*in-
verse*
func-
tion
of
 f ,

3.2.

COM-

PO-

see

SP-

INTER-

ITION

AND

INVERSE

3.2 Composition and in- verse

Composing

func-

tions

is

the

pro-

cess

of

ap-

ply-

ing

func-

tions

in

se-

quence,

one

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ter

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other.

You

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math-

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mat-

ics.

It

is

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mat-

CHAPTER
3.
FUNCTIONS

i-
cal
con-
cept
and
we
can
study
it
even
be-
fore
ad-
mit-
ting
any
no-
tion
of
num-
ber
or
mea-
sure-
ment
etc.
It
will
play
an
im-
por-
tant
role
in
our
study
of
groups
in
the
next
chap-
ter.

Definition

3.11.

Let
 $f :$
 $X \rightarrow$
 Y
and
 $g :$
 $Y \rightarrow$
 Z
be

3.2.
 COM-
 PO-
 TWO
 SY-
 M-
 METR-
 ION
 TIONS
 AND
 INVERSE
 that

the
 co-
 domain
 of
 f
 is
 also
 the
 do-
 main
 of
 g).
 Then
 we
 can
 form
 a
 new
 func-
 tion

$$g \circ f : X \rightarrow Z,$$

defined
 by
 the
 for-
 mula

$$(g \circ f)(x) = g(f(x)).$$

We
 can
 re-
 fer
 to
 $g \circ$
 f
 as
 g
com-
posed
with
 f ,
 or
 more
 in-
 for-
 mally,
 g
af-
ter
 f .

CHAPTER
3.
FUNCTIONS

Example

3.3.

Con-
sider
the
real-
valued
func-
tions
 f
and
 g
de-
fined
by
the
for-
mu-
las

$$f(x) = x^2 - 3 \text{ and } g(x) = 2x + 3.$$

Then
we
can
find
for-
mu-
las
defin-
ing
the
com-
po-
si-
tions
as
fol-
lows

$$(f \circ g)(x) = f(g(x)) = (2x + 3)^2 - 3 = 4x^2 + 12x + 6,$$

and

$$(g \circ f)(x) = g(f(x)) = 2(x^2 - 3) + 3 = 2x^2 - 3.$$

Note
that
these
for-
mu-
las
are
dif-
fer-
ent.
In

3.2. COM- PO- SITION AND INVERSE

both
com-
po-
si-
tions
are
de-
fined,
com-
po-
si-
tion
is
not
com-
mu-
ta-
tive,
i.e.

$$f \circ g \neq g \circ f,$$

although
in
spe-
cial
cases
they
may
be
equal.

Example

3.4.

Con-
sider
the
real-
valued
func-
tions
 g, h, r
de-
fined
by
the
for-
mu-
las

$$g(x) = 2x - 7, \quad h(x) = x^2 - 1 \text{ and } r(x) = e^{2x}.$$

Obtain
for-

CHAPTER
3.
FUNCTIONS

mu-
las
for
each
of
the
fol-
low-
ing
com-
po-
si-
tions,

1.
 $g \circ$
 $h,$
2.
 $h \circ$
 $r,$
3.
 $g \circ$
 $h \circ$
 $r,$
4.
 $g \circ$
 $g,$
5.
 $g \circ$
 $g \circ$
 $g,$
6.
 $r \circ$
 $r.$

Definition

3.12

(Iden-
tity
func-
tion).

The
iden-
tity
func-
tion
id
on
the
do-
main
 X
is
the

3.2.

COM-

PO-

func-

tion

tion

AND

$\text{id} : X \rightarrow X$,

defined

by

the

for-

mula

$$\forall x \in X \quad \text{id}(x) = x.$$

We

shall

speak

of

the

iden-

tity

func-

tion

in

dif-

fer-

ent

con-

texts

with-

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ily

spec-

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fy-

ing

the

do-

main.

The

do-

main

can

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de-

ter-

mined

from

the

spe-

cific

con-

text.

CHAPTER
3.
FUNCTIONS

The
iden-
tity
func-
tion
has
the
prop-
erty
that
when
com-
posed
with
an-
other
func-
tion,
it
leaves
the
other
func-
tion
un-
changed,
i.e.

$$f \circ \text{id} = \text{id} \circ f = f.$$

Sometimes
the
ef-
fect
of
a
func-
tion
can
be
re-
versed,
as
in
the
fol-
low-
ing
def-
i-
ni-
tion.

Definition

3.13

(In-
verse

3.2.

COM-

PO-

func-

tion

AND

INVERSE

$X \rightarrow$

Y

be

a

func-

tion

that

is

bi-

jec-

tive,

i.e.

is

in-

jec-

tive

and

sur-

jec-

tive.

Then

there

ex-

ists

a

func-

tion

$f^{-1} : Y \rightarrow X,$

called

the

in-

verse

of

$f,$

which

has

the

prop-

erty

that

$f^{-1} \circ f = f \circ f^{-1} = \text{id}.$

Or

in

other

words,

$\forall x \in X \quad f^{-1}(f(x)) = x,$

and

$\forall y \in Y \quad f(f^{-1}(y)) = y.$

CHAPTER
3.
FUNCTIONS
Example

3.5.

Given

a
func-
tion
de-
fined
by

a
for-
mula

it
is
of-
ten
pos-
si-
ble

to
ob-
tain
the
for-
mula

for
the
in-
verse
func-
tion
(if

it
ex-
ists).

Sup-
pose
that
the
real-
valued
func-
tion

f
is
de-
fined
by
the
for-
mula

$$f(x) = 2x - 5.$$

We
form
the

3.2. COM- PO- SITION AND INVERSE

and
ad-
just
it
so
that
 x
be-
comes
the
sub-
ject
of
the
equa-
tion,
which
is

$$x = \frac{y + 5}{2}.$$

This
now
gives
us
the
cor-
rect
for-
mula
for
the
in-
verse
func-
tion,
 f^{-1} ,
which
is
de-
fined
by

$$f^{-1}(y) = \frac{y + 5}{2}.$$

In
terms
of
graphs,
there
is

CHAPTER
3.
FUNCTIONS

a
nice
re-
la-
tion-
ship
be-
tween
the
graph
of
a
real-
valued
func-
tion
and
that
of
its
in-
verse.
The
graph
of
the
in-
verse
func-
tion
can
be
ob-
tained
by
re-
flect-
ing
the
graph
of
the
orig-
i-
nal
func-
tion
in
the
di-
ag-
o-
nal
line
in
the

3.2.
 COM-
 PO-
 SI-
 TION
 de-
 AND
 fined
 INVERSE
 by
 $y =$
 x .
 This
 has
 the
 ef-
 fect
 of
 in-
 vert-
 ing
 the
 re-
 la-
 tion-
 ship
 be-
 tween
 the
 do-
 main
 and
 co-
 domain.

Example

3.6.

Fig-
 ure
 3.2
 shows
 the
 graphs
 of
 the
 square
 func-
 tion
 $s :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}^+$
 de-
 fined
 by
 $s(x) =$
 x^2
 and
 square
 root
 func-
 tion

CHAPTER
3.
FUNCTIONS

$r :$
 $\mathbb{R}^+ \rightarrow$
 $\mathbb{R}^+$
de-
fined
by
 $r(x) =$
 $\sqrt{x},$
along
with
the
dot-
ted
line
show-
ing
the
axis
of
re-
flec-
tion
 $y =$
 $x.$

The
next
re-
sult
re-
lates
the
in-
verse
of
the
com-
po-
si-
tion
of
two
func-
tions
to
the
com-
po-
si-
tions
of
their
in-
verses.
It
should

3.2.
COM-
PO-

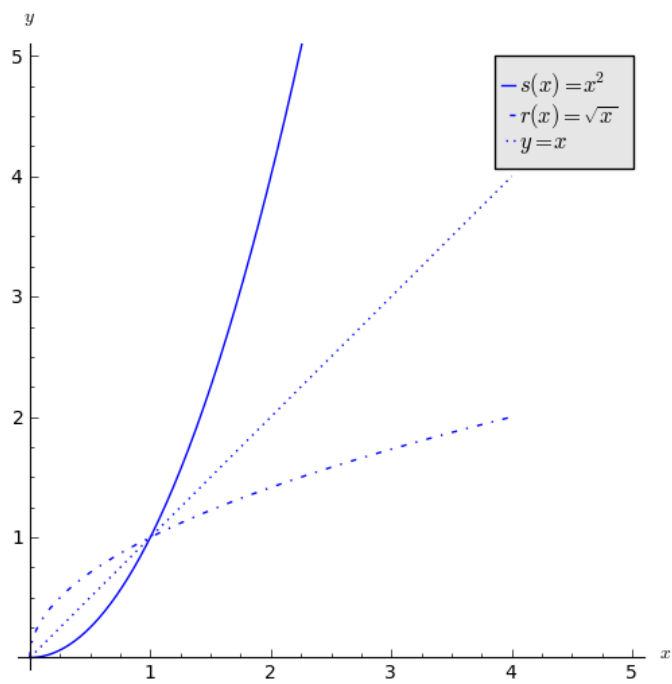


Figure 3.2:
The
square
func-
tion
and
its
in-
verse

re-
mind
you
of
a
sim-
i-
lar
re-
sult
about
ma-
trix
in-
verses.

Theorem
3.1.
*Sup-
pose
that*

CHAPTER
3.
FUNCTIONS

$f:$
 $X \rightarrow$
 Y

and

$g:$
 $Y \rightarrow$
 Z

are
two
bi-
jec-
tive
func-
tions.

Then
the
com-
po-
si-
tion

$g \circ$
 f
is
also
bi-
jec-
tive
and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

3.3 Rational func- tions and their par- tial frac- tion ex- pan- sions

Rational
func-
tions
are
those
func-
tions
that
are
quo-

3.3. RA- TIO- NALS OF FUNC- TIONS AND THEIR FAC- TIAL FRAC- TION EXPANSIONS

ra-
tio-
nal
func-
tion
 r ,
is
a
real-
valued
func-
tion
of
the
form

$$r = \frac{p}{q},$$

where
the
func-
tions
 p
and
 q
are
poly-
no-
mi-
als.

The
treat-
ment
of
ra-
tio-
nal
func-
tions
(for
in-
stance
when
in-
te-
grat-
ing

CHAPTER
3.
FUNCTIONS
them)
is
of-
ten
made
eas-
ier
by
re-
expressing
them
in
terms
of
par-
tial
frac-
tions.
This
is
a
tech-
nique
for
ex-
press-
ing
a
ra-
tio-
nal
func-
tion
as
a
sum
of
ra-
tio-
nal
func-
tions
of
a
very
sim-
ple
type.
An
ex-
am-
ple
of
a
par-
tial

3.3.
RA-
TIO-
NAL
FAC-
TION
FUNC-
TIONS
AND
THEIR
PAR-
TIAL
FRAC-
TION
EXPANSIONS
func-
tion
is:

$$\begin{aligned}\frac{x}{x^3 + x^2 + 2x - 4} &= \frac{x}{(x + 2)^2(x - 1)} \\ &= \frac{-1/9}{x + 2} + \frac{2/3}{(x + 2)^2} + \frac{1/9}{x - 1},\end{aligned}$$

where
the
fi-
nal
form
is
called
the
par-
tial
frac-
tion
ex-
pan-
sion
of
the
orig-
i-
nal
ra-
tio-
nal
func-
tion.
The
par-
tial
frac-
tions
tech-
nique
re-
lies
on
the
fol-
low-

CHAPTER
3.
FUNCTIONS

ing
fact
about
poly-
no-
mi-
als
with
real
co-
ef-
fi-
cients.

Theorem
3.2

(Poly-
no-
mial
Fac-
tori-
sa-
tion).

Let
p
be
a
poly-
no-
mial
with
real
co-
ef-
fi-
cients
and
lead-
ing
co-
ef-
fi-
cient
1,
i.e.

$$p(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_2x^2 + a_1x + a_0.$$

Then
p(x)
fac-
tor-
izes
into
a
prod-
uct

3.3. RA- TIO- NAL POLY- FUNG- TIONS AND THEIR PAR- TIAL FRAC- TION EXPANSIONS

cients,
each
of
which
is
ei-
ther
a
lin-
ear
fac-
tor
of
the
form
 $(x - a)^m$,
where
 $a \in \mathbb{R}$
and
 $m \in \mathbb{N}$
or
a
quadratic
fac-
tor
of
the
form
 $(x^2 + ax + b)^m$,
where
 $a, b \in \mathbb{R}$
and
 $m \in \mathbb{N}$.
A
fac-
tor
 $(x - a)^m$

CHAPTER
3.
FUNCTIONS

cor-
re-
sponds
to
a
real
root
a
of
p
of
mul-
ti-
plic-
ity
m
and
a
quadratic
type
fac-
tor
 $(x^2 +$
 $ax +$
 $b)^m$
cor-
re-
sponds
to
a
pair
of
com-
plex
con-
ju-
gate
roots
of
p
of
mul-
ti-
plic-
ity
m.

Proof.
Omitted.
□

Below
is
a
com-
plete

3.3.
 RA-
 TIO-
 NAL
 de-
 scrip-
 tion
 of
 the
 method
 of
 their
 partial
 frac-
 tion
 ex-
 pan-
 sion
 of
 a
 ra-
 tio-
 nal
 func-
 tion.
 Fol-
 low-
 ing
 that
 we
 give
 some
 ex-
 am-
 ples.
 You
 may
 find
 it
 eas-
 ier
 to
 fol-
 low
 the
 ex-
 am-
 ples
 first,
 re-
 fer-
 ring
 back
 to
 the
 gen-
 eral

CHAPTER
3.
FUNCTIONS
de-
scrip-
tion
of
the
method
as
needed.

The Gen- eral Method

So
sup-
pose
that
 $r =$
 $\frac{p}{q}$
is
a
ra-
tio-
nal
func-
tion.
To
find
the
par-
tial
frac-
tion
ex-
pan-
sion
we
first
fac-
tor-
ize
 $q(x)$
in
the
way
de-
scribed
by
the
above
the-
o-
rem.
For
each

3.3.
RA-
TIO-
NAL
FAC-
TOR
FUNC-
TIONS
AND
THEIR
FORM
EXPAN-
SION
EXPANSIONS

we
in-
clude
the
terms

$$\frac{\alpha_1 x + \beta_1}{x^2 + ax + b} + \frac{\alpha_2 x + \beta_2}{(x^2 + ax + b)^2} + \cdots + \frac{\alpha_m x + \beta_m}{(x^2 + ax + b)^m},$$

in
the
par-
tial
frac-
tion
ex-
pan-
sion.
The
real
num-
bers
 α_i, β_i
will
be
de-
ter-
mined
later.

For
each
fac-
tor
of
the
form
 $(x - a)^m$
of
 $q(x)$
we
in-
clude
the
terms

$$\frac{\alpha_1}{x - a} + \frac{\alpha_2}{(x - a)^2} + \cdots + \frac{\alpha_m}{(x - a)^m},$$

CHAPTER
3.
FUNCTIONS

in
the
par-
tial
frac-
tion
ex-
pan-
sion.
The
real
num-
bers
 α_i
will
be
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later.

If
the
de-
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mer-
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tor
poly-
no-
mial
 $p(x)$
equals
or
ex-
ceeds
the
de-
gree
of
the
de-
nom-
i-
na-
tor
poly-
no-
mial
 $q(x)$
by
 m ,
where
 $m \geq$

3.3.

RA-

TIO-

QUAL

FUNCTIONS

AND

THEIR

PARTIAL

FRAC-

TION

$\lambda_0 + \lambda_1 x + \cdots + \lambda_m x^m,$

EXPANSIONS

the

par-

tial

frac-

tion

ex-

pan-

sion,

the

num-

bers

λ_i

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Following

these

steps

will

give

you

the

par-

tial

frac-

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pan-

sion

for

$r(x)$.

To

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mine

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ues

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CHAPTER
3.
FUNCTIONS
fi-
cients
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tial
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pan-
sion
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 $q(x)$
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de-
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na-
tors
of
the
par-
tial

3.3.
 RA-
 TIO-
 NAL
 FUNC-
 TIONS
 AND
 THEIR
 PAR-
 TIAL
 FRAC-
 TION
 OF
 EXPANSIONS
 to
 the
 poly-
 no-
 mial
 $q(x)$
).
 This
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 leave
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 equa-
 tion
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 als
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 fi-
 cients.
 We
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 co-
 ef-
 fi-
 cients
 now

CHAPTER
3.
FUNCTIONS

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pow-
ers
of
 x
from
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sides
of
the
equa-
tion.
(Re-
mem-
ber
that
two
poly-
no-
mi-
als
are
equal
if
and
only
if
their
cor-
re-
spond-
ing
co-
ef-
fi-
cients
are
equal).

Example

3.7.
Here
are
two
ex-
am-
ples
that

3.3.
RA-
TIO-
NAL
show
the
FUNC-
TIONS
method
work
AND
img
THEIR
PAR-
TIAL
FRAC-
TION¹
An
EXPANSIONS
am-

¹www.wolframalpha.com

will
pro-
vide
the
par-
tial
frac-
tion
ex-
pan-
sion
of
any
ra-
tio-
nal
func-
tion
that
is
sub-
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ted
to
it.
In
ad-
di-
tion
by
click-
ing
the
show
steps
link
in
the
out-
put
it
will
dis-
play
a
step-
by-
step
deriva-
tion
of
the
ex-
pan-
sion

CHAPTER
3.
FUNCTIONS

ple
with
only
sim-
ple
fac-
tors
in
the
de-
nom-
i-
na-
tor

$$\begin{aligned}\frac{3x+1}{x^3-3x+2} &= \frac{3x+1}{(x-1)^2(x+2)} \\ &= \frac{\alpha_1}{x-1} + \frac{\alpha_2}{(x-1)^2} + \frac{\beta}{x+2}.\end{aligned}$$

Here
we
have
fac-
tor-
ized
the
de-
nom-
i-
na-
tor
and
formed
the
par-
tial
frac-
tion
ex-
pan-
sion.
To
de-
ter-
mine
the
par-
tial
frac-
tion
ex-
pan-
sion
we
mul-
ti-

3.3.

RA-

TIO-

NAL ply

FUNC both

TIONS sides

AND by

THEIR $(x -$

PAR- $1)^2(x +$

TIAL- $2)$

FRAC to

TION get

EXPANSIONS

$$\begin{aligned} 3x + 1 &= \alpha_1(x - 1)(x + 2) + \alpha_2(x + 2) + \beta(x - 1)^2 \\ &= (\alpha_1 + \beta)x^2 + (\alpha_1 + \alpha_2 - 2\beta)x + (-2\alpha_1 + 2\alpha_2 + \beta). \end{aligned}$$

Equating

co-

ef-

fi-

cients

from

both

sides

of

the

poly-

no-

mial

gives

us

the

si-

mul-

ta-

ne-

ous

equa-

tions

$$\alpha_1 + \beta = 0,$$

$$\alpha_1 + \alpha_2 - 2\beta = 3,$$

$$-2\alpha_1 + 2\alpha_2 + \beta = 1.$$

Solving

these

equa-

tions

leads

to

the

so-

lu-

tion

$$\alpha_1 = \frac{5}{9}, \alpha_2 = \frac{4}{3}, \beta = \frac{-5}{9}.$$

So

the

CHAPTER
3.
FUNCTIONS

partial
fraction
expansion
of
our
original
rational
function
is

$$\frac{3x+1}{x^3-3x+2} = \frac{5/9}{x-1} + \frac{4/3}{(x-1)^2} - \frac{5/9}{x+2}.$$

2.

An
example
with
a
quadratic
type
factor.

$$\begin{aligned} \frac{5x}{x^3-x^2-x-2} &= \frac{5x}{(x^2+x+1)(x-2)} \\ &= \frac{\alpha_1+\beta x}{x^2+x+1} + \frac{\alpha_2}{x-2} \end{aligned}$$

Multiplying
across
by
(x^2+
 $x+$
 1)($x-$
 2)
gives

$$5x = (\alpha_1+\beta x)(x-2)+\alpha_2(x^2+x+1).$$

Multiplying
out
the
right
hand
side
and
then

3.3.
RA-
TIO-
NAL equat-
FUNC-
TION^{ing}
AND^{co-}
THEIR^{ef-}
PAR-^f
TIAL^{icients}
FRAC-^{of}
TION^{the}
EXPANSIONS^{two}
Poly-
no-
mi-
als
leads
to
the
val-
ues

$$\alpha_1 = \frac{5}{7}, \beta = \frac{-10}{7}, \alpha_2 = \frac{10}{7}.$$

And
so
the
par-
tial
frac-
tion
ex-
pan-
sion
is

$$\frac{5x}{x^3 - x^2 - x - 2} = \frac{\frac{5}{7} - \frac{10}{7}x}{x^2 + x + 1} + \frac{\frac{10}{7}}{x - 2}.$$

Exercises

3.1.

These
prob-
lems
re-
quire
the
ap-
pli-
ca-
tion
of
var-
i-
ous
con-
cepts
and
re-
sults

CHAPTER
3.
FUNCTIONS
from
this
chap-
ter.
We
can
use
these
as
a
ba-
sis
for
fur-
ther
dis-
cus-
sion
in
the
tu-
to-
ri-
als.

1. Consider
the
func-
tions
 u, v, w
de-
fined
by
 $u(x) =$
 $x +$
 $1,$
 $v(x) =$
 $x^2 +$
 1
and
 $w(x) =$
 $\sin x.$
Give
a
defin-
ing
for-
mula
for
the
fol-
low-
ing
func-
tions

3.3.
RA-
TIO-
NAL a)
FUNC- $u \circ$
TIONS v
AND b)
THEIR $v \circ$
PAR- w
TIAL c)
FRAC- $v \circ$
TION $u \circ$
EXPANSIONS

2.
Let
 S ,
 R ,
 Exp
and
 One
de-
note
the
func-
tions
de-
fined
by
 $S(x) =$
 x^2 ,
 $R(x) =$
 $\sqrt{x}$,
 $\text{Exp}(x) =$
 e^x
and
 $\text{One}(x) =$
1.
Con-
sider
the
func-
tions
 f
and
 g
de-
fined
by

$$f(x) = \left(3e^{2x^2}\right)^{\frac{1}{2}} \text{ and } g(x) = 2e^{(1+\sqrt{x})^2}.$$

Express
 f
and
 g
as
suit-
able

CHAPTER
3.
FUNCTIONS

com-
po-
si-
tions
of
com-
bi-
na-
tions
of
the
func-
tions
 S ,
 R
and
Exp.

3. Consider
the
func-
tion
 f, g, h
de-
fined
by
the
fol-
low-
ing
for-
mu-
lae:

$$f(x) = 3x+1, \quad g(x) = \frac{x}{x-3}, \quad h(x) = \frac{x+1}{x-1}.$$

What
are
ap-
pro-
pri-
ate
do-
mains
and
co-
domains
for
these
func-
tions?
Find
for-
mu-
lae
for

3.3.
RA-
TIO-
NAL their
in-
FUNC-
TIONS
AND
THEIR
PAR-
TIAL
FRAC-
TION
EXPANSIONS
domains
of
these
new
func-
tions.

4.
Let
the
func-
tions
 $\lambda :$
 $\mathbb{R} \rightarrow$
 $(-1, \infty)$
be
de-
fined
by

$$\lambda(t) = e^{\frac{t}{2}} - 1.$$

Find
the
for-
mula
for
the
func-
tion
 λ^{-1} ,
stat-
ing
its
do-
main
and
co-
domain.

5.
Let
the
func-
tions
 $\phi :$
 $(0, \infty) \rightarrow$

CHAPTER

3.

FUNCTIONS

$(-1, \infty)$

and

$\psi :$

$\mathbb{R} \rightarrow$

$(0, \infty)$

be

de-

fin-

ed

by

$$\phi(x) = x^2 - 1, \quad \psi(x) = e^{2x}.$$

Find

the

def-

i-

ni-

tions

of

the

func-

tions

$\phi \circ$

ψ

and

$(\phi \circ$

$\psi)^{-1}$

and

con-

firm

that

$$(\phi \circ \psi)^{-1} = \psi^{-1} \circ \phi^{-1}$$

in

this

case.

6.

Obtain

par-

tial

frac-

tion

ex-

pan-

sions

of

the

fol-

low-

ing

ra-

tio-

nal

func-

tions

3.4.

IN-

FI-

NITE a)

SETS

AND

CARDINALITY

$$\frac{2s+2}{s^2-2s-8},$$

b)

$$\frac{s^2+s-3}{s^2+s},$$

c)

$$\frac{s^2+3}{s^3-5s^2+s-5}.$$

3.4 Infinite

sets

and

car-

di-

nal-

ity

We

in-

clude

here

a

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tion

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in-

tro-

duces

some

im-

por-

tant

con-

cepts

that

are

con-

nected

to

deep

is-

sues

in

set

the-

ory,

logic

and

the

CHAPTER
3.
FUNCTIONS

foundations and philosophy of mathematics. The main question is how to speak precisely about the size of a set. We shall see how the answer to this question uses the concepts of injectivity and surjectivity of maps

3.4.
IN-
FI-
NITE
SETS
AND
CARDINALITY

**The
num-
ber
of
el-
e-
ments
in
a
set**

Intuitively,
to
work
out
the
size
of
a
set
we
should
sim-
ply
count
the
num-
ber
of
el-
e-
ments
that
it
has.
So
some
triv-
ial
ex-
am-
ples
are

- the
set
 $\{a, b, c\}$
has
size
3,

CHAPTER
3.
FUNCTIONS

(as-
sum-
ing
 a, b, c
are
dis-
tinct
ob-
jects),

- the
set
of
in-
te-
gers
 $\{10, 11, 12, 13, 14\}$
has
size
5,
- the
set
 $\{7\}$
has
size
1,
- the
empty
set
 $\{\}$
has
size
0.

So
surely,
given
any
set,
we
can
sim-
ply
count
the
num-
ber
of
el-
e-
ments
that
it

3.4.
IN-
FI-
nite
has
and
SETS
the
AND
we
CARDINALITY
know
its
size
and
can
com-
pare
sets
in
this
way.
This
ap-
proach
is
per-
fectly
cor-
rect
and
would
be
the
whole
story
if
one
was
only
con-
sid-
er-
ing
fi-
nite
sets.
But
in
math-
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mat-
ics
we
soon
en-
counter
many
in-
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est-
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CHAPTER
3.
FUNCTIONS

finite
sets
and
would
like
to
speak
pre-
cisely
about
their
sizes
and
be
able
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com-
pare
them.

The
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con-
cept
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size
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tor.
The

3.4.
 IN-
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 NITE
 SETS
 AND
 CARDINALITY
 of
car-
di-
nal-
ity
 is
 used.

Definition

3.14

(Car-
 di-
 nal-
 ity).
 We
 say
 that
 two
 sets
 A
 and
 B
 have
 the
 same
 car-
 di-
 nal-
 ity
 if
 there
 ex-
 its
 a
 bi-
 jec-
 tion
 $\phi :$
 $A \rightarrow$
 B .

Notice
 that
 this
 def-
 i-
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 tion
 does
 not
 ex-
 plic-
 itly

CHAPTER
3.
FUNCTIONS

tell
you
what
the
car-
di-
nal-
ity
(or
size)
of
a
set
is,
rather
it
sup-
plies
a
cri-
te-
rion
(the
ex-
is-
tence
of
a
bi-
jec-
tion)
for
when
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re-
gard
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sets
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the
same
car-
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To
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this
def-
i-
ni-
tion

3.4.
IN-
FL-
think
NITE
of
SETS
the
AND
PRO-
CARDINALITY
cess

of
count-
ing
the
el-
e-
ments
of
a
fi-
nite
set
with.
To
count
the
el-
e-
ments
of
a
set
is
re-
ally
to
set
up
a
pair-
ing
(or
bi-
jec-
tion)
of
the
el-
e-
ments
of
the
set
with
the
count-
ing
num-
bers
 $1, 2, 3, \dots, n,$
for

CHAPTER
3.
FUNCTIONS

some

ap-
pro-
pri-
ate
value

n .

Or

in
other
words,
when

we
count
the

el-
e-
ments

in
a
fi-
nite

set

X

we
are
re-
ally
con-
struct-
ing

a
bi-
jec-
tion

$$X \rightarrow \{1, 2, 3, \dots, n\}.$$

The
beauty

of
Can-
tor's
def-

i-
ni-
tion

is
that

it
sim-
ply

ex-
tends

this
idea

to

3.4.
 IN-
 FI-
 NITE
 SETS
 AND
 CARDINALITY
 finite
 and
 in-
 fi-
 nite.

We
 will
 write
 $|A| =$
 $|B|$
 when
 A
 and
 B
 have
 the
 same
 car-
 di-
 nal-
 ity.
 The
 ex-
 is-
 tence
 of
 in-
 jec-
 tions
 be-
 tween
 sets
 also
 has
 im-
 pli-
 ca-
 tions
 for
 car-
 di-
 nal-
 ity.
 If
 there
 ex-
 ists
 an
 in-
 jec-
 tion
 $\psi :$

CHAPTER
3.
FUNCTIONS

$A \rightarrow$
 B

then

we

can

say

that

the

car-

di-

nal-

ity

of

A

is

no

big-

ger

than

B

and

write

$|A| \leq$

$|B|$.

Sim-

i-

larly

if

there

ex-

ists

a

sur-

jec-

tion

$\rho :$

$A \rightarrow$

B

we

can

say

that

the

car-

di-

nal-

ity

of

A

is

no

less

than

B

and

write

3.4. FINITE SETS AND CARDINALITY

Example 3.8.

We can see this definition operating when we assess the cardinalities of the familiar basic number systems.

- the natural numbers
 $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- the integers
 $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

CHAPTER
3.
FUNCTIONS

- the
ra-
tio-
nals
 $\mathbb{Q} =$
 $\{\frac{p}{q} :$
 $p, q \in$
 $\mathbb{Z}, q \neq$
 $0\}$
- the
real
num-
bers
 $\mathbb{R}$,
the
set
of
points
on
the
num-
ber
line
(or
*con-
tinuum*).

See
the
lec-
tures
for
de-
tails
of
these
ex-
am-
ples.

The
ba-
sic
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3.4.
 IN-
 FI-
 NITE
 SETS
 AND
 CARDINALITY
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 ing
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 te-
 rion
 for
 an
 in-
 fi-
 nite
 set.

Definition

3.15

(In-
 fi-
 nite
 set).

A
 set
 A
 is
*in-
 fi-
 nite*
 if
 there
 ex-
 ists
 a
 bi-
 jec-
 tion

$$\phi : A \rightarrow C \subsetneq A,$$

from
 A
 to
 a
 proper
 sub-
 set
 of
 it-
 self.

CHAPTER
3.
~~FUNCTIONS~~
**The
Con-
tin-
uum
Hy-
poth-
e-
sis**

Sets
hav-
ing
the
same
car-
di-
nal-
ity
as
 $\mathbb{N}$
are
called
*count-
able*.
Hav-
ing
proved
that
 $\mathbb{R}$
is
un-
count-
able
the
ques-
tion
arises,
are
there
sets
with
in-
ter-
me-
di-
ate
size?
That
is,
does
there
ex-
ists
a
set

3.4. IN- FI- NITE SETS AND CARDINALITY

$|\mathbb{N}| < |X| < |\mathbb{R}|?$

The
Con-
tin-
uum
Hy-
poth-
e-
sis
(CH)
is
that
there
is
no
such
set
X.

This
may
seem
like
a
straight-
for-
ward
enough
ques-
tion,
at
first.
But
it
has
turned
out
to
be
ex-
tremely
elu-
sive
and
has
deep
con-
nec-
tions
to
the

CHAPTER
3.
FUNCTIONS
foun-
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tions
and
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los-
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phy
of
math-
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ics.

**Status
of
CH
com-
pared
with
the
rest
of
set
the-
ory**

Zermelo-
Fraenkel
set
the-
ory
and
the
Ax-
iom
of
Choice
(ZFC)
is
the
com-
monly
used
ax-
iomatic
ba-
sis
for
set-
theory
(and
hence
most
math-

3.4. IN- FI- NITE SETS AND CARDINALITY

- Kurt Gödel proved in 1940 that CH can-not be dis-proved within ZFC.

- Paul Co-hen proved in 1963 that CH can-not be proved within ZFC.

These re-sults show that CH is in-de-pen-dent of ZFC, or rel-a-tively con-sis-tent with

CHAPTER
3.
FUNCTIONS

ZFC,
mean-
ing
that
as-
sum-
ing
CH
(or
its
con-
verse
 $\neg$ CH)
will
not
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duce
any
con-
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dic-
tions
into
set
the-
ory
that
were
not
there
to
be-
gin
with.

Attitudes

to
the
ques-
tion
"Does
there
ex-
ists
a
set
 X
with
 $|N| <$
 $|X| <$
 $|\mathbb{R}|$?"
re-
flect
dif-
fer-
ent

3.4.
IN-
FI-
NITE
SETS
AND
CARDINALITY
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Resources

Here
are
some
links
to
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sources
on
these
and
re-
lated
top-
ics.

- Hotel
Hilbert
([https://
www.
youtube.
com/
watch?
v=
revY6cmX5m8](https://www.youtube.com/watch?v=revY6cmX5m8)),
an
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mo-
spheric
BBC/Open
Uni-
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sity
drama
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CHAPTER

3.

FUNCTIONS

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of
rooms.

- To
In-
fin-
ity
and
Be-
yond
([http://
www.
bbc.
co.
uk/
programmes/
b00qszch](http://www.bbc.co.uk/programmes/b00qszch))
a
BBC
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- Read
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about
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3.4.

IN-

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SETS^{Stan-}

AND^{ford}

89^{En-}

CARDINALITY

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([http:](http://plato.stanford.edu/entries/set-theory/)

[//](http://plato.stanford.edu/entries/set-theory/)

[plato.](http://plato.stanford.edu/entries/set-theory/)

[stanford.](http://plato.stanford.edu/entries/set-theory/)

[edu/](http://plato.stanford.edu/entries/set-theory/)

[entries/](http://plato.stanford.edu/entries/set-theory/)

[set-theory/](http://plato.stanford.edu/entries/set-theory/))

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([http:](http://plato.stanford.edu/entries/philosophy-mathematics/)

[//](http://plato.stanford.edu/entries/philosophy-mathematics/)

[plato.](http://plato.stanford.edu/entries/philosophy-mathematics/)

[stanford.](http://plato.stanford.edu/entries/philosophy-mathematics/)

[edu/](http://plato.stanford.edu/entries/philosophy-mathematics/)

[entries/](http://plato.stanford.edu/entries/philosophy-mathematics/)

[philosophy-mathematics/](http://plato.stanford.edu/entries/philosophy-mathematics/)).